

Competitive Exam Practice 4 — Answer Key

Expression Notation, Recursion, and Tower of Hanoi

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

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- Q1 (A).** Only operators ever wait on the conversion stack (operands go straight to the output; parentheses are markers, not content) — so an operator stack is the one essential structure. An operand stack is what postfix *evaluation* needs, not conversion; a parse tree is sufficient but not necessary. Text and options confirmed against the GATEOverflow V2 p.231 heading and ExamSIDE's stacks-and-queues listing (2026-09-09); the year tag differs across those two sources, so none is claimed.
- Q2 (A) `abc**def^^-`.** Scan left to right: `a` out; `+` waits; `b` out; `*` outranks `+` so it waits too; `c` out; `-` flushes `*` then `+` (both outrank or tie it left-associatively); `d` out; the first `^` waits above `-`; the second `^` does *not* pop the first (right-associative: pop only on strictly greater precedence); `f` out; end flushes `^^ -`. Option (B) wrongly pops the first `^` (treats it as left-associative, giving $d^{(ef)}$ the wrong grouping $(d^e)^f$); (C) misplaces `b`; (D) is a prefix string, not postfix. Confirmed against the SlideShare scan of the official GATE CS 2004 solved paper (options verbatim), the gatecse.in GATE-2004 key (Q38 = a), and exact re-derivation by execution (2026-09-09).
- Q3 (A) 6, 1 (top first).** Trace: push 8, 2, 3; `^` pops 3, 2 giving $2^3 = 8$, stack [8, 8]; `/` gives $8/8 = 1$, stack [1]; push 2, 3; the first `*` pops 3, 2 giving $2 \times 3 = 6$, stack [1, 6] — top element 6, next 1. (B)/(C) are intermediate values from mis-grouping; (D) reverses the order. Confirmed against the GATEOverflow page for GATE CSE 2007 Q38 (accepted answer A), ExamSIDE's GATE-2007 page, the Electrical4U GATE-2007 solutions, and exact re-trace by execution (2026-09-09).
- Q4 Condition: `n >= 1` (accept `n > 0`, `n != 0`); first call: `move(n-1, A, C, B)`; second call: `move(n-1, B, A, C)`.** The skeleton prints the n -th-disk move *inside* the `if`, so the condition must admit every $n \geq 1$ and use $n = 0$ as the silent base case (no `else` exists, hence `n == 1` is wrong — it would skip every larger tower). With `printf` moving `A -> C`, `A` is the source and `C` the destination, so the first recursive call parks $n - 1$ disks on `B`: `move(n-1, A, C, B)`, and the second restacks them from `B` onto `C`: `move(n-1, B, A, C)`. Skeleton text from GATEOverflow V2 p.310 (single source for the year); completion derived deterministically from the deck's Hanoi algorithm.
- Q5 (A) `**ab-cd`.** Reverse and swap parens: $(d-c)*(b+a)$; its postfix is `dc-ba**`; reverse back: `**ab-cd`. (B) swaps the operand pairs; (C) puts `+` outside (forgets the reversal step); (D) flips a sign. Confirmed by execution (2026-09-09).

Q6 (B) $\mathbf{ab*cd-+}$. Scan right to left: $\mathbf{d}$, $\mathbf{c}$ combine under $-$ first ($\mathbf{cd-}$); $\mathbf{b}$, $\mathbf{a}$ combine under $*$ ($\mathbf{ab*}$); $+$ joins them: $\mathbf{ab*cd-+}$. (A) is the deck's own answer to the *different* prefix $\mathbf{**ab-cd}$ — the planted near-miss; (C) misplaces the last operator; (D) reverses operand order. Confirmed by execution (2026-09-09).

Q7 63. $T(1) = 1$, $T(n) = 2T(n-1) + 1$ gives $T(n) = 2^n - 1$; $2^6 - 1 = 63$.

Q8 (B) $(63, 6)$. $M = 2^6 - 1 = 63$ moves; the longest chain of simultaneously live calls never exceeds the disk count, so $D = 6$. (A) counts 2^n instead of $2^n - 1$; (C) confuses total calls with live depth; (D) is the $n = 5$ instance from the deck's exit ticket.